
Provably Protecting Fine-Tuned LLMs from Training Data Extraction while Preserving Utility

Tom Segal¹ Yuval Elovici¹ Asaf Shabtai¹

Abstract

Fine-tuning large language models (LLMs) on sensitive datasets raises privacy concerns, as training data extraction (TDE) attacks can expose highly confidential information. Existing defenses against such attacks either lack formal privacy guarantees or incur substantial utility degradation. We observe that fine-tuning induces widespread probability shifts, yet preserving only a small subset of influential token-level deviations is sufficient; the remaining shifts can be aggressively smoothed with minimal impact on utility. Motivated by this insight, we propose SCP- Δ_r , a Near Access Freeness (NAF)-based algorithm that operates on relative probabilities and explicitly smooths low-impact tokens using a base model. SCP- Δ_r achieves orders-of-magnitude better theoretical bounds than existing NAF based methods and provides strong empirical protection against TDE attacks with minimal performance loss.

Code: https://github.com/ppol/scp_dr

1. Introduction

Large language models (LLMs) are known to memorize parts of their training data, enabling training data extraction (TDE) attacks that recover verbatim sequences from black-box access to a model. Such attacks have been demonstrated against commercial models such as ChatGPT and Gemini (Carlini et al., 2021; Nasr et al., 2025). This risk is amplified by fine-tuning, which is routinely performed on small, high-value, and often private datasets (Patil & Gudivada, 2024; J et al., 2024).

Existing defenses face a fundamental trade-off. Empirical mitigations such as deduplication or modified losses (Kand-

pal et al., 2022; Hans et al., 2024) can reduce memorization in practice but offer no formal guarantees and are increasingly circumvented by new attacks (Nasr et al., 2025). Conversely, differential privacy (DP) provides rigorous protection but often incurs substantial utility loss or computational overhead when applied to LLM fine-tuning (Blanco-Justicia et al., 2022; Du et al., 2025). As a result, there is currently no practical defense with meaningful theoretical guarantees against TDE for fine-tuned LLMs.

In this work, we bridge this gap by developing a theoretically grounded yet practical protection mechanism based on Near Access Freeness (NAF) (Vyas et al., 2023). NAF constructs a model that is provably close, under a divergence measure, to multiple constituent models, guaranteeing safety if at least one constituent is safe for a given input. While NAF has been applied to copyright protection, access control, and privacy (Li et al., 2024; Kalai et al., 2025; Segal et al., 2025), its implications for TDE have not been previously studied. In particular, we show that the widely used NAF based CP- Δ algorithm can yield weak guarantees in realistic fine-tuning settings, since memorization induces sharp probability spikes on a small number of tokens.

Sparse relative probability shifts. Our key observation is that fine-tuning concentrates the most significant changes in model outputs on a relatively small subset of tokens, even though absolute token probabilities may vary widely due to normalization. As a result, fine-tuned models can appear far apart under divergences defined over raw probabilities, yielding weak CP- Δ guarantees, despite differing meaningfully, i.e., in ways that affect the generated token, on only a small set of tokens. We formalize this phenomenon via the Sparse Relative Probability Shift (SpaRPS) property, an idealized abstraction of this structural sparsity. We observe that fine-tuned LLMs may not naturally satisfy this property due to widespread shifts in per-token probabilities induced by fine-tuning. We conjecture that most shifts do not lead to meaningful changes in generated outputs, and thus SpaRPS can be enforced via a smoothing mechanism while preserving downstream performance.

SCP- Δ_r : enforcing SpaRPS via relative probabilities and smoothing. Building on this insight, we propose SCP-

¹Department of Software and Information Systems Engineering, Ben-Gurion University of the Negev, Beer Sheva, Israel. Correspondence to: Tom Segal <tomsega@post.bgu.ac.il>.

Δ_r , a NAF-based algorithm designed to explicitly exploit and enforce SpaRPS for TDE protection.

SCP- Δ_r combines two complementary components:

1. **Relative probabilities.** Following Segal et al. (2025), we operate on relative probability distributions (RPDs) rather than raw probabilities. This removes global normalization effects and yields dramatically smaller NAF bounds when probability shifts are sparse in relative space, as predicted by SpaRPS.
2. **Base-model smoothing.** To ensure that SpaRPS holds in practice despite widespread probability shifts, we introduce a smoothing mechanism that aligns most token-level relative probabilities with those of a shared base model, preserving only the most informative deviations. This explicitly enforces SpaRPS while incurring minimal utility loss.

Together, these changes enable SCP- Δ_r to achieve orders-of-magnitude smaller NAF bounds than CP- Δ .

From NAF guarantees to training data extraction protection. Crucially, NAF guarantees do not automatically imply protection against training data extraction. We show that CP- Δ bounds may allow large information gain for a TDE adversary. In contrast, we prove that under reasonable and empirically satisfied assumptions, SCP- Δ_r yields small bounds on the information gain of a TDE adversary. Attempts to derive comparable guarantees for CP- Δ result in bounds that do not meaningfully constrain TDE.

Empirical evaluation. We evaluate SCP- Δ_r against CP- Δ and other NAF-based methods using both existing and newly proposed TDE attacks. Our results show that SCP- Δ_r achieves the strongest protection against TDE attacks, with no observable utility degradation. In contrast, we find that previously proposed NAF-based methods can be highly vulnerable to TDE attacks.

Contributions.

1. We introduce SpaRPS, a property capturing the idealized sparsity structure of fine-tuning-induced relative probability shifts, and show that fine-tuned models can be smoothed to satisfy this property while preserving performance.
2. We propose SCP- Δ_r , a practical NAF-based defense that operates on relative probability distributions and enforces SpaRPS via base-model smoothing, achieving strong theoretical and empirical protection against TDE with minimal utility degradation.
3. We establish the first formal connection between NAF guarantees and training data extraction resistance, showing that existing NAF-based methods may fail to meaningfully constrain TDE, while SCP- Δ_r yields dramatically smaller extraction bounds under standard assumptions.
4. We demonstrate the first practical extraction attacks against NAF-based methods, and show that previously proposed methods remain practically vulnerable, whereas SCP- Δ_r achieves substantially stronger empirical and theoretical resistance to these attacks.

In the remainder of the paper, we first discuss relevant background on TDE and NAF. We then introduce the SpaRPS property and empirically study its applicability to LLMs. Next, we present SCP- Δ_r together with its smoothing mechanism for enforcing SpaRPS, and derive provable guarantees connecting CP-style algorithms to TDE. Finally, we evaluate SCP- Δ_r against existing NAF-based methods using both adaptive and non-adaptive extraction attacks.

2. Background and Related Work

2.1. Preliminaries

We use the following notation. Let T denote a training (or fine-tuning) algorithm that maps a dataset S to a conditional generative model $p(\cdot|\cdot)$. For an input x (also referred to as a prompt) and output token y , $p(y|x)$ denotes the probability assigned to y given x . We occasionally write p to denote a distribution instead of a model. We write $\Delta(p||q)$ for a divergence between distributions p and q , and $D(p, q)$ for a symmetric distance; divergences need not be symmetric. All models and probability distributions are enforced to assign probability of at least e^{-20} to all outputs.

2.2. Training Data Extraction

LLMs exhibit a phenomenon known as "eidetic memorization," where they learn to reproduce verbatim sequences from their training corpora to minimize cross-entropy loss on "long-tail" data (Carlini et al., 2021; Tirumala et al., 2022). This creates significant privacy vulnerabilities, as adversaries can extract sensitive information, including Personally Identifiable Information (PII) and proprietary code, by interacting with the model's black-box API (Carlini et al., 2021; Nasr et al., 2025). While safety alignment (e.g., RLHF) was initially thought to mitigate this, recent "divergence attacks" can force models to deviate from their answer style, increasing the rate of training data leakage by orders of magnitude (Nasr et al., 2025).

Training data extraction of fine-tuned models. Fine-tuning introduces increased risks as it typically involves

smaller, high-value datasets where the probability shift relative to the base model serves as a strong membership signal. Empirical studies indicate that fine-tuning can amplify PII extraction rates compared to pre-trained baselines (Miresghallah et al., 2022b; Akkus et al., 2025). Crucially, parameter-efficient methods are not immune; the "LoRA-Leak" framework demonstrates that Low-Rank Adaptation (LoRA) updates still yield high membership inference success (AUC ≈ 0.77) by leveraging the pre-trained base model as a reference (Ran et al., 2025; Marinelli & Eckhoff, 2026).

Defenses against training data extraction. Defensive strategies are generally categorized into empirical mitigations and provable guarantees. In-practice methods include training data deduplication and PII scrubbing, though recent work shows that context-based attacks can still reconstruct redacted entities with high accuracy (Lee et al., 2022; Lukas et al., 2023). Training-time interventions like "Goldfish Loss," which randomly masks tokens during loss computation, effectively reduce verbatim memorization but lack formal privacy bounds (Hans et al., 2024). Conversely, Differential Privacy (DP), typically implemented via DP-SGD, provides rigorous mathematical guarantees by clipping gradients and adding noise, but it frequently incurs substantial utility costs and computational overhead that hinder widespread adoption (Dwork, 2006; Abadi et al., 2016; Du et al., 2025). In contrast to all above approaches, we base our method on NAF and achieve both practicality and theoretically grounded protection.

2.3. NAF and CP- Δ

NAF constructs a model $p(\cdot|x)$ that remains close, under a divergence Δ , to a set of constituent models $p_1(\cdot|x), \dots, p_m(\cdot|x)$. The premise is that if, for a given input x , at least one constituent model behaves safely, then an aggregate model that is sufficiently close to all constituents should also behave safely. NAF has been applied to copy-right protection, access control, poisoning robustness, and privacy (Vyas et al., 2023; Li et al., 2024; Kalai et al., 2025; Segal et al., 2025).

Definitions and theorems of this subsection follow Vyas et al. (2023).

Definition 2.1 (k -NAF). A model $p(\cdot|x)$ is k_x -NAF on input x with respect to models $p_1(\cdot|x), \dots, p_m(\cdot|x)$ and divergence Δ if, for all $i \leq m$,

$$\Delta(p(\cdot|x) \| p_i(\cdot|x)) \leq k_x.$$

If $\Delta(p(\cdot|x) \| p_i(\cdot|x)) = k_x$ for all i , we call $p(\cdot|x)$ k_x -NAF-tight. If $k_x \leq k$ for all x , the model is called k -NAF.

A common instantiation uses the max divergence $\Delta(p\|q) := \max_y \log \frac{p(y)}{q(y)}$, which implies pointwise probability control (Vyas et al., 2023):

$$p(y|x) \leq e^{k_x} p_i(y|x) \quad \text{for all } i, y.$$

CP- Δ . A standard NAF method for $m = 2$ is CP- Δ , which takes a pointwise minimum of two distributions and renormalizes:

Definition 2.2 (CP- Δ). Let p, q be distributions over Y . Define

$$\text{CP-}\Delta(p, q)(y) := \frac{\min(p(y), q(y))}{Z},$$

$$Z := \sum_{y \in Y} \min(p(y), q(y)).$$

Theorem 2.3 (CP- Δ bound). For any input x , CP- $\Delta(p(\cdot|x), q(\cdot|x))$ is k_x -NAF-tight with respect to $p(\cdot|x)$ and $q(\cdot|x)$ under max divergence, where

$$k_x = D_m(p(\cdot|x), q(\cdot|x)), \quad D_m(p, q) := \log \frac{1}{1 - \text{TV}(p, q)}$$

$$\text{and } \text{TV}(p, q) := \frac{1}{2} \sum_y |p(y) - q(y)|.$$

Remark 2.4. The quantity $D_m(p, q)$ can be numerically large: $\text{TV}(p, q)$ may approach 1 even when p and q differ substantially on only a small subset of tokens, yielding guarantees that are not meaningful in practice (Cohen, 2025).

To illustrate how memorization-induced probability spikes affect CP-style bounds, consider the following example:

Example 2.1. Let p and q be probability distributions over $Y = \{0, \dots, 9\}$ such that $q(y) = 0.1$ for all $y \in Y$, while $p(0) = 0.91$ and $p(y) = 0.01$ for all $y \neq 0$ (See Table 1).

The distributions in Example 2.1 illustrate a common memorization pattern in fine-tuned LLMs. Here, p concentrates most of its mass on a single memorized token, while q remains balanced. As a result, the pointwise minimum $\min(p(y), q(y))$ is small for most y , driving $\text{TV}(p, q)$ close to 1 and yielding a large CP- Δ bound. This behavior shows how sparse probability spikes, which are characteristic of memorization, can cause CP-style bounds based on raw probabilities to become numerically large.

Table 1. Example distributions p and q and their CP-style aggregations. Applying CP to raw probabilities (P) yields a highly imbalanced distribution compared to applying CP to relative probabilities (R). Bold columns indicate the probability representation on which each aggregation operates.

	p		q		CP- Δ		CP- Δ_r	
	P	R	P	R	P	R	P	R
0	0.91	58	0.1	1	0.53	7.9	0.15	1.5
1-9	0.01	0.64	0.1	1	0.05	0.79	0.09	0.96

2.4. Relative Probabilities and $\text{CP-}\Delta_r$

To reduce sensitivity to probability spikes, we follow (Segal et al., 2025) and operate on *relative probability distributions* (RPDs). While (Segal et al., 2025) did not formulate their method within the NAF framework, it can be viewed as applying the CP aggregation rule to relative probabilities rather than to raw probabilities; we refer to this variant as $\text{CP-}\Delta_r$. Definitions and theorems of this subsection follow Segal et al. (2025), rephrased in NAF notation.

Definition 2.5 (RPD). Let q be a distribution over Y and let $n := |Y|$. Define the *typical probability* tq and the corresponding *relative probability distribution* (RPD) rq by

$$tq := \exp\left(\frac{1}{n} \sum_{y \in Y} \log q(y)\right), \quad rq(y) := \frac{q(y)}{tq}.$$

Conceptually, raw probabilities are sensitive to global normalization effects. Relative probabilities instead isolate changes in the model’s relative token preferences, which more directly reflect meaningful behavioral differences between models. Mathematically, RPDs do not sum to 1; instead, they satisfy $\sum_y \log rq(y) = 0$, reducing sensitivity to probability spikes and often yielding smaller bounds in practice (Segal et al., 2025). This construction addresses the failure mode illustrated in Example 2.1, as illustrated in Table 1.

Definition 2.6 ($\text{CP-}\Delta_r$). Let p, q be distributions over Y . Define

$$\text{CP-}\Delta_r(p, q)(y) := \frac{\min(rp(y), rq(y))}{Z},$$

where $Z := \sum_{y \in Y} \min(rp(y), rq(y))$.

Theorem 2.7 ($\text{CP-}\Delta_r$ bound). For any input x , $\text{CP-}\Delta_r(p(\cdot|x), q(\cdot|x))$ is k_x -NAF-tight with respect to $p(\cdot|x)$ and $q(\cdot|x)$ under the relative max-divergence $\Delta_r(p||q) := \max_y \log \frac{rp(y)}{rq(y)}$, with

$$k_x = D_r(p(\cdot|x), q(\cdot|x)), \quad D_r(p, q) := \frac{1}{2n} \sum_y \log \frac{rp(y)}{rq(y)}.$$

As discussed in Section 4.1, the bound in Theorem 2.7 is dramatically smaller in practice than the bound in Theorem 2.3 (Figure 1b).

2.5. Other NAF Variations

CP-style algorithms have also been analyzed under alternative divergences such as KL (Vyas et al., 2023). Recent work observed that sequence generation under $\text{CP-}\Delta_{\text{KL}}$ may drift toward one constituent model, compromising safety, and proposed fusing strategies to mitigate this effect (Abad et al., 2025). As discussed in subsection 5.3, such fusing does not address our adaptive extraction attacks.

3. The SpaRPS Property

In this section, we introduce the SpaRPS property and discuss its applicability to fine-tuned LLMs. First, we give theoretical justification for using relative probabilities, which SpaRPS operates on.

3.1. Why Relative Probabilities

Table 1 illustrates a key failure mode of divergences defined on raw probabilities: even if the difference between two distributions is caused by a single spike, normalization effects can make $p(y)$ and $q(y)$ far apart for all y .

Concretely, suppose p and q arise from unnormalized “scores” that differ by an unknown multiplicative constant C . A natural way to estimate this constant is to choose C that best aligns p and q in log-space on average. The next theorem shows that, under squared error in log-space, the optimal scaling is exactly the ratio of typical probabilities, which is equivalent to using RPDs.

Theorem 3.1. Let p and q be distributions over Y with $p(y), q(y) > 0$ for all $y \in Y$. Then

$$\arg \min_{C>0} \sum_{y \in Y} \log^2 \left(C \frac{p(y)}{q(y)} \right) = \frac{tq}{tp}.$$

The proof follows from a standard least-squares argument and is provided in Appendix B.1.

3.2. SpaRPS

We are now ready to define the SpaRPS (Sparse Relative Probability Shift) property.

Definition 3.2 (SpaRPS). Let p and q be distributions over Y . We say that p is m -SpaRPS with respect to q if there exists a set $A \subseteq Y$ with $|A| = m$ and $\alpha > 0$ such that

$$p(y) = \alpha q(y) \quad \forall y \in Y \setminus A.$$

We emphasize that SpaRPS is intended as an idealized structural abstraction rather than a literal property expected to hold exactly for real fine-tuned LLMs without intervention.

Proposition 3.3. Let $n := |Y|$. If p is m -SpaRPS with respect to q and $\forall y \in A : \log \frac{p(y)}{q(y)} \leq \log M$, then

$$D_r(p, q) \leq \frac{m}{n} (\log M + |\log \alpha|).$$

For $m = 20$ and $n = 32000$ (as used in our evaluation), and assuming $\log M + |\log \alpha| \leq 40$ (which follows from all probabilities being lower bounded by e^{-20}), we obtain $D_r(p, q) \leq 0.025$. A proof is provided in Appendix B.2.

By Proposition 3.3 and Theorem 2.7, fine-tuned LLMs that satisfy SpaRPS are close in terms of relative probability

distributions and therefore provide a small $\text{CP-}\Delta_r$ bound. However, due to widespread probability shifts induced by fine-tuning, it is unlikely in practice that $p(\cdot|x)$ is exactly m -SpaRPS with respect to $q(\cdot|x)$. In particular, the ratio $\frac{p(y|x)}{q(y|x)}$ is unlikely to equal a single constant α for all but m outputs simultaneously. We therefore introduce *Expected SpaRPS* as a more realistic relaxation.

Definition 3.4 (Expected SpaRPS). Let T be a randomized training algorithm and let S_p, S_q be datasets. For each prompt $x \in X$, define the random conditional models $p(\cdot|x) := T(S_p)(x)$ and $q(\cdot|x) := T(S_q)(x)$, where the randomness is over T . Let $n := |Y|$.

We say that (S_p, S_q) satisfies (m, B, C) -Expected SpaRPS on X if there exists a subset $X' \subseteq X$ with $|X'| \geq C|X|$ such that for every $x \in X'$ there exists a set $A_x \subseteq Y$ with $|A_x| = m$ satisfying

$$|\mathbb{E}[\log rp(y|x)] - \mathbb{E}[\log rq(y|x)]| \leq B \quad \forall y \in Y \setminus A_x.$$

Expected SpaRPS can be viewed as a PAC-style relaxation of SpaRPS, in which the proportionality condition is allowed to hold approximately (via B) and for most prompts (via C), under the randomness of the training algorithm. Ideally, $B \approx 0$ and $C \approx 1$.

Figure 1a illustrates the relationship between the Expected SpaRPS parameters m , B , and C in a practical fine-tuning setting. The datasets S_p and S_q contain 3000 and 2900 training examples respectively, with $S_q \subset S_p$, and the SpaRPS analysis evaluates prompts $X = S_p \setminus S_q$. We estimate $\mathbb{E}[\log rp(y|x)]$ and $\mathbb{E}[\log rq(y|x)]$ by averaging log relative probabilities over 16 independent fine-tunes (different seeds) of LLaMA2-7B on the *MathAbstracts* dataset (see Section 5.1). We find that even Expected SpaRPS does not hold in practice. As shown in Figure 1a, even when allowing $(m = 1024)$ unconstrained shifts, $(1 - C = 10\%)$ of prompts still exhibit large deviation ($B \approx 2.1$). This suggests that SpaRPS must be explicitly enforced if it is to serve as the basis for practical TDE guarantees.

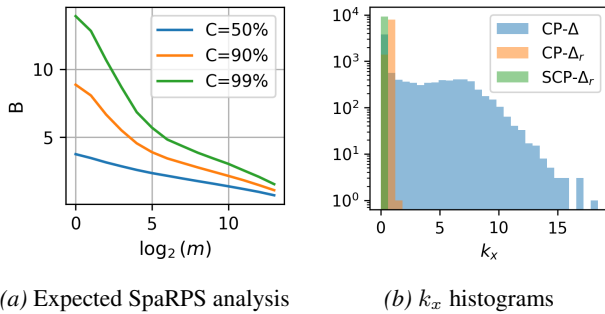


Figure 1. Expected SpaRPS and k_x evaluated using the *MathAbstracts* dataset (See 5.1). (a) Expected SpaRPS B as a function of $\log_2(m)$ for different C values. (b) k_x bounds log-scale histograms for different CP style algorithms.

4. SCP- Δ_r

We observed that SpaRPS and Expected SpaRPS do not realistically hold in practice. We conjecture, however, that LLMs behave as though they satisfy SpaRPS, in the sense that retaining only a small number of relative probability shifts would not meaningfully affect model behavior. To enforce SpaRPS and enable its use in our theoretical analysis, we introduce the S (smoothing) component of $\text{SCP-}\Delta_r$, which enforces the stronger, non-expected form of SpaRPS (Definition 3.2) by replacing most relative probabilities with those of a shared base model b prior to aggregation. Our experiments (Section 5.4) validate our conjecture by showing that the smoothing mechanism of $\text{SCP-}\Delta_r$ does not degrade utility.

4.1. Base-Model Smoothing

Let $b(\cdot|x)$ denote a fixed *base* model (e.g., the pre-fine-tuning model). Given an RPD rp and a base RPD rb , we define smoothing with parameter m via $S_m(rp, rb)$, which preserves the relative probabilities of m selected tokens and sets all remaining tokens to match the base model, up to a global rescaling.

Formally, let $I \subseteq Y$ be a token set of size m (chosen by a scoring rule defined below). Define $S_m(rp, rb)$ by

$$S_m(rp, rb)(y) = \begin{cases} \beta rp(y), & y \in I, \\ \beta rb(y), & y \notin I, \end{cases}$$

where $\beta > 0$ is chosen s.t. $\sum_{y \in Y} \log S_m(rp, rb)(y) = 0$ (i.e., the result is an RPD). We use $\tilde{rp}$ to denote a smoothed RPD. Note that for small m , β is close to 1 (see Appendix B.6).

We choose I as the m tokens with largest score

$$\text{score}_{p,b}(y) := p(y|x) \cdot \log \frac{rp(y|x)}{rb(y|x)},$$

This score is the per-token contribution to the relative KL divergence: $\Delta_{\text{KL}}(p(\cdot|x) \| b(\cdot|x)) = \sum_{y \in Y} \text{score}_{p,b}(y)$. Selecting the top- m tokens by $\text{score}_{p,b}$ bounds the potential utility degradation caused by smoothing (see Appendix A). **Remark 4.1.** $S_m(rp, rb)$ is $2m$ -SpaRPS with respect to $S_m(rq, rb)$ for any RPDs rp, rq and rb . A proof is provided in Appendix B.3.

Algorithm 1 describes $\text{SCP-}\Delta_r$ with a smoothing parameter m . The resulting model is equivalent to $\text{CP-}\Delta_r(\tilde{p}, \tilde{q})$, where $\tilde{p}$ and $\tilde{q}$ are the probability distributions induced by the relative probability distributions $\tilde{rp}$ and $\tilde{rq}$. As a result, any theoretical guarantees for $\text{CP-}\Delta_r$ apply directly to $\text{SCP-}\Delta_r$. Figure 1b shows that the resulting k_x values for $\text{SCP-}\Delta_r$ are substantially smaller than those of $\text{CP-}\Delta_r$, which in turn are smaller than those of $\text{CP-}\Delta$. At the 99th percentile, the

Algorithm 1 SCP- Δ_r (for a fixed input)

Input: Distributions b, p, q and smoothing level m .

Return: A distribution r .

- 1: $rb \leftarrow \text{RPD}(b); rp \leftarrow \text{RPD}(p); rq \leftarrow \text{RPD}(q)$.
- 2: $\hat{r}p \leftarrow S_m(rp, rb); \hat{r}q \leftarrow S_m(rq, rb)$.
- 3: $u(y) \leftarrow \min(\hat{r}p(y), \hat{r}q(y))$ for all $y \in Y$.
- 4: **return** $r(y) \leftarrow u(y) / \sum_{y' \in Y} u(y')$.

corresponding values are 0.004, 0.98, and 11.6, respectively, reflecting orders-of-magnitude improvements when moving from CP- Δ to CP- Δ_r , and from CP- Δ_r to SCP- Δ_r . Note that CP- Δ bounds k_x under the standard max-divergence on raw probabilities, whereas CP- Δ_r and SCP- Δ_r use relative max-divergence; we compare these quantities as they play the same role in the NAF guarantee.

Importantly, achieving these improvements does not incur substantial computational overhead: Training cost scales approximately linearly with the amount of training data, which is partitioned across the two constituent models. During inference, the three forward passes are independent and can therefore be executed in parallel on a single GPU. Moreover, the aggregation operations are lightweight relative to the forward passes themselves, and based on prior empirical evaluations of CP-style algorithms, the resulting wall-clock overhead is expected to be modest in practice (Abad et al., 2025).

4.2. SCP- Δ_r TDE Protection

We now discuss SCP- Δ_r protection against TDE. Theorems 4.4 and 4.5 below are stated for CP- Δ_r , but apply directly to SCP- Δ_r by replacing relative probability distributions with their smoothed counterparts, i.e., $rq \rightarrow \hat{r}q$.

TDE framework. Typically, TDE attacks consist of two stages. First, the adversary queries the target model to obtain candidate training examples, and then applies a membership inference (MI) attack to predict membership for each candidate (Carlini et al., 2021; Nasr et al., 2025). We assume the adversary possesses prefixes of targeted training examples and aims to extract the next token for each prefix. We further assume the adversary has black-box access to the target model’s output probabilities and white-box access to a reference model trained in similar manner and on similar data as the target model, which is standard in prior work on MI (Mireshghallah et al., 2022a; Galende et al., 2025).

Definition 4.2 (Black-box TDE Game). Let $S_e = \{s_1, \dots, s_n\} \subset S$ and let r_1 be an LLM trained on S . Let $X_e = \{x_1, \dots, x_n\}$ be such that, for all i , x_i is a prefix of s_i . A TDE adversary ALG is given X_e , a reference LLM r_0 , and black-box access to r_1 , allowing it to query r_1 with any x and observe $r_1(y|x)$ for all $y \in Y$. The adversary outputs

a prediction y_i for each x_i or abstains. The success of ALG is measured by its accuracy on predicted tokens as a function of coverage (the fraction of non-abstained predictions), as standard in selective classification (Geifman & El-Yaniv, 2017).

Remark 4.3. We restrict the adversary to predicting specified training examples rather than allowing it to generate arbitrary examples. This is because some training examples are inherently easier to extract than others, and an unrestricted adversary would simply focus on the easiest cases. Our formulation instead enables us to reason about the extractability of a given training example.

Likelihood ratio attack (LiRA). Without loss of generality, we assume that the adversary first picks a candidate y_i for each x_i (the generation stage) and then decides whether to keep it or abstain (the MI stage). While an optimal strategy for the full TDE game is intractable (Sablayrolles et al., 2019), it has been shown that under reasonable assumptions (discussed in Appendix C.1), the optimal strategy for the MI stage is thresholding the likelihood ratio $\frac{r_1(y_i|x_i)}{r_0(y_i|x_i)}$ (LiRA) (Sablayrolles et al., 2019; Mireshghallah et al., 2022a; Galende et al., 2025). Although LiRA is derived for the MI stage, the same ratio also characterizes the information gain obtained from observing a candidate token during generation. Consequently, uniformly bounding this ratio limits the maximum advantage attainable by any candidate-generation strategy, and therefore constrains both the generation and MI stages of TDE attacks.

In the simplified case that $r_0 = q$, standard NAF guarantees yield a direct bound on the likelihood ratio. Indeed, for $r_1 = \text{CP-}\Delta(p, q)$, we have $\frac{r_1(y|x)}{q(y|x)} \leq e^{k_x}$, as explained in subsection 2.3. For CP- Δ_r , we present the following result:

Theorem 4.4 (CP- Δ_r LiRA bound for $r_0 = q$). *Let $r_1 = \text{CP-}\Delta_r(p, q)$, $r_0 = q$ and k_x the bound in Theorem 2.7 then*

$$\frac{r_1(y|x)}{r_0(y|x)} = \frac{r_1(y|x)}{q(y|x)} \leq \frac{tr_1(x)}{tq(x)} \cdot e^{k_x} =: e^{t_x},$$

where $tq(x)$ and $tr_1(x)$ denote the typical probabilities of $q(\cdot|x)$ and $r_1(\cdot|x)$.

Proof and empirical analysis of Theorem 4.4 are provided in Appendix B.4 and D.3 respectively.

For $r_0 \neq q$, we assume that $r_0 = \text{CP-}\Delta_r(p_0, q)$ for some model p_0 . We note that the more similar r_0 is to r_1 (apart from its training set not including the target set S_e), the better for the adversary, so this is the “worst case”.

Theorem 4.5 (LiRA bound for $r_0 = \text{CP-}\Delta_r(p_0, q)$). *Let $r_i = \text{CP-}\Delta_r(p_i, q)$ for $i \in \{0, 1\}$, and k_x the corresponding bound from Theorem 2.7 for r_1 , then*

$$\frac{r_1(y|x)}{r_0(y|x)} \leq \max \left(1, \frac{rq(y|x)}{rp_0(y|x)} \right) \cdot \frac{tr_1(x)}{tr_0(x)} \cdot e^{k_x} =: e^{v_x(y)}$$

A proof for Theorem 4.5 is provided in Appendix B.5.

Remark 4.6. While typical probabilities are affected by probability spikes, even if p_1 exhibits such spikes, the aggregated model r_1 need not, due to the CP aggregation with q . As a result, the ratios $\frac{tr_1(x)}{tr_0(x)}$ and $\frac{tr_1(x)}{tq(x)}$ appearing in Theorems 4.4 and 4.5 are expected to be small for most prompts in practice.

Remark 4.7 (Theorem 4.5 variation for CP- Δ). Let $r_i = \text{CP-}\Delta(p_i, q)$ for $i \in \{0, 1\}$, and k_x the corresponding bound from Theorem 2.3, then $\frac{r_1(y|x)}{r_0(y|x)} \leq \max(1, \frac{q(y|x)}{p_0(y|x)}) \cdot e^{k_x}$. Note that for CP- Δ , k_x is empirically large compared to CP- Δ_r and SCP- Δ_r (Figure 1b).

Proposition 4.8 ($v_x(y)$ is small for smoothed (rare) tokens). For SCP- Δ_r , suppose that y is smoothed in both $\tilde{r}q$ and $\tilde{r}p_0$. Then, under the parameters of Proposition 3.3, $\frac{\tilde{r}q(y|x)}{\tilde{r}p_0(y|x)} \leq 1.026$, which implies

$$e^{v_x(y)} \leq 1.026 \cdot \frac{tr_1(x)}{tr_0(x)} \cdot e^{k_x} \leq 1.053 \cdot \frac{tr_1(x)}{tr_0(x)}.$$

Notably, under the conditions of Propositions 3.3 and 4.8, together with Remark 4.6, all terms contributing to the bound $e^{v_x(y)}$ are small. A proof and illustration of Proposition 4.8 are provided in Appendix B.6 and D.4, respectively.

Figure 2 illustrates empirical histograms of the bounds implied by Theorem 4.5 in a rare vs typical token scenarios. Here, $p_i = T(S_i)$ and $q = T(S_q)$. The datasets S_1 , S_0 , and S_q contain 3000, 2900, and 3000 training examples respectively, with $S_0 \subset S_1$ and $S_1 \cap S_q = \emptyset$. The results in Figure 2 indicate that SCP- Δ_r substantially improves the extraction bound relative to CP- Δ_r and CP- Δ , especially for rare tokens, as Proposition 4.8 suggests.

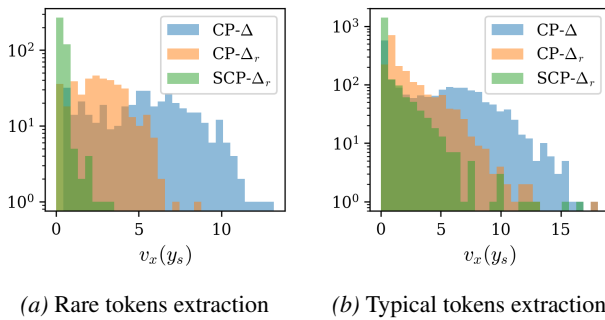


Figure 2. Log-scale histograms of Theorem 4.5 bound $v_x(y_s)$, evaluated at the token y_s corresponding to the extraction target s . (a) Extraction of rare tokens (Canaries, see 5.3). (b) Extraction of tokens from the *MathAbstracts* dataset (see 5.1).

5. Evaluation

5.1. Experimental Setup

We evaluate the same datasets as Abad et al. (2025), in addition to a synthesized *Names2IDs* dataset, for a total of four datasets: *MathAbstracts* (Zhang et al., 2025b) (paper titles and abstracts), *StoryTelling* (Fan et al., 2018) (story topics and stories), *CodeInstructions* (Morales, 2023) (instructions and Python solutions), and *Names2IDs* (synthetic names paired with 10-digit IDs). For *CodeInstructions* we fine-tune StarCoder-7B (Li et al., 2023); for the remaining datasets we use LLaMA2-7B (Touvron et al., 2023).

Unless stated otherwise, we fine-tune two constituent models p and q on disjoint partitions of the training set, using 3000 examples per partition (6000 total), for 50 epochs to encourage memorization, following Abad et al. (2025). The smoothing parameter is set to $m = 10$ by default (see Appendix C.3 on this choice). All experiments were conducted on an NVIDIA RTX 6000 GPU with 48 GB of memory. Additional training details are provided in Appendix D.1.

Compared models. We compare CP- Δ , CP- Δ_{KL} (Vyas et al., 2023), CP-Fuse (Abad et al., 2025), CP- Δ_r (Segal et al., 2025), and our method SCP- Δ_r . We evaluate two variants of SCP- Δ_r : SCP- Δ_r -b uses the pre-fine-tuning model as the base model, while SCP- Δ_r -c uses a constant logits vector as the base model, obtained by averaging the pre-fine-tuning model’s logits over 100 held-out examples. As an additional baseline, we report results for *No CP*, which directly uses the model trained on the first partition (i.e., p). For reference, we report the validation-set results of *No CP*, together with their standard error (STE).

5.2. Utility Evaluation

Since the fine-tuning datasets of the two constituent models originate from the same source distribution, their output distributions are expected to remain similar under typical usage. Consequently, aggregation is not expected to meaningfully degrade utility. This observation was also validated empirically in prior NAF-based works (Vyas et al., 2023; Abad et al., 2025; Segal et al., 2025). Nevertheless, SCP- Δ_r introduces an additional smoothing component, motivating a dedicated evaluation of its utility impact. We report next-token accuracy on held-out validation sets for all experiments, and additionally evaluate downstream reasoning and instruction-following performance with and without smoothing (Appendix D.7). We also derive utility degradation bounds for SCP- Δ_r in Appendix A.

5.3. Extraction of NAF-Protected LLMs’ Training Data

In this subsection, we empirically evaluate NAF-protected models against TDE attacks. While our theoretical analysis

Table 2. Full results table. Best values are in bold. Second best values are underlined.

	NAS ↓	Canary ↓		PII ↓		TTE ↓						Utility ↑		
	All	Math		IDs		Math		Story		Code		Math	Story	Code
	Mean	Mean	95%	AEL	FER	AUC	ACC	AUC	ACC	AUC	ACC	ACC	ACC	ACC
No CP	.99	28	31	7.4	.73	.94	.84	.98	.92	.96	.84	.47	.37	.67
CP- Δ	.47	8.8	17	2.4	.15	.90	.69	.96	.80	.89	.76	.50	.41	.70
CP- Δ_{KL}	.76	23	33	4.1	.34	.93	.77	.97	.86	.88	.80	.51	.42	.70
CP-Fuse	.50	3.8	7.1	4.1	.34	.93	.77	.97	.86	.88	.80	.48	.38	.68
CP- Δ_r	.22	3.6	8.0	1.9	.06	.80	.50	.86	.58	.80	.61	.50	.40	.69
SCP- Δ_r -c	.17	<u>.16</u>	<u>.34</u>	<u>.18</u>	<u>.05</u>	<u>.79</u>	<u>.49</u>	.83	<u>.59</u>	<u>.81</u>	<u>.61</u>	.49	.40	.69
SCP- Δ_r -b	.11	.14	.32	.43	<.001	.76	.46	<u>.84</u>	.60	.78	.60	<u>.50</u>	.39	.72
No CP-val	.002	1.4	3.6	.12	1e-10	.66	.38	.71	.45	.71	.51	.47	.37	.67
STE	-	.11	.90	.01	-	.01	.01	.01	.01	.02	.01	.01	.01	.01

considers optimal LiRA attacks, our empirical attacks do not explicitly implement LiRA. Nevertheless, they can be interpreted as instantiating LiRA with a near-ideal reference model, as discussed in Appendix C.2. We note that when the adversary is restricted (e.g., when the reference model is far from ideal), previously proposed MIA attacks have been shown to outperform LiRA (Fu et al., 2023; Zhang et al., 2025a). This occurs because the effectiveness of LiRA degrades under restricted adversaries. In our setting, however, we are unaware of attacks stronger than LiRA, which is optimal under certain assumptions (Appendix C.1).

The canary method (Carlini et al., 2019). The canary method consists of inserting sequences of random words to the training set and then evaluating the model memorization of them. We use the *exposure score* defined by Carlini et al. (2019). An exposure score w indicates that the canary ranks among the 2^{-w} most probable sequences according to the model. Our canaries consist of 3 randomly generated words. We inserted each canary 3 times into the *MathAbstracts* training set (all in the same partition) and trained the models for 5 epochs. Appendix D.5 further reports results for canaries inserted 1 or 10 times.

PII extraction. We use the *Names2IDs* dataset to simulate extraction of personally identifiable information (PII). Given a prompt of the form “Name: <name>, ID:”, the adversary queries the model, records the next generated token, appends it to the prompt, and repeats adaptively. We report the average extracted length (AEL) and the full ID extraction rate (FER).

Token-by-token extraction (TTE). TTE attempts to reconstruct training examples in two adaptively repeated stages. First, the adversary queries the model using a prefix of a training example and records the next generated token. Second, the adversary predicts whether this token corresponds to the true label by thresholding its probability under the model. We report single token extraction success using the selective classification framework (allowing ab-

stention), measuring the accuracy-coverage AUC (standard in selective classification (Geifman & El-Yaniv, 2017)) and the accuracy of the attack when the adversary always accepts the generated token (ACC). We evaluate TTE on the *MathAbstracts*, *StoryTelling*, and *CodeInstruction* datasets.

Importantly, the fusing mechanism of CP-Fuse has no effect on the first generated token (Abad et al., 2025). Since both TTE and our PII extraction method are adaptive and rely only on the first generated token for each prompt, CP-Fuse is effectively reduced to CP- Δ_{KL} in this setting.

5.4. Results

We report the main results and ablation studies below. Additional results, including experiments on LoRA fine-tuned LLMs, are presented in Appendix D.

Main results. Table 2 summarizes security under the canary method and PII and TTE extraction attacks, together with next-token accuracy on held-out data. We also report the average normalized attack score (NAS) across the three attacks. Specifically, each attack score (column) is normalized to $[0, 1]$ (excluding the STE row), and NAS is computed by averaging the normalized scores across columns, with TTE scores scaled by a factor of $1/3$ to balance the contribution of all attack types. SCP- Δ_r -b provides the strongest protection under the canary method and PII extraction, and achieves security comparable to CP- Δ_r on TTE, consistent with Proposition 4.8. SCP- Δ_r -c performs similarly to SCP- Δ_r -b, except on PII extraction, which we analyze further in Appendix C.4. Utility remains comparable across CP-style methods, with only minor differences. Notably, non-RPD-based CP-style algorithms remain highly vulnerable to our attacks, with NAS values ranging from 0.47 to 0.76. In contrast, SCP- Δ_r -b achieves an NAS of 0.11, substantially closer to the reference baseline (No CP-val, NAS = 0.002) and substantially improving over CP- Δ_r (NAS = 0.22).

Effect of smoothing level. Figure 3a shows the effect of the smoothing parameter m on the canary score for SCP- Δ_r .

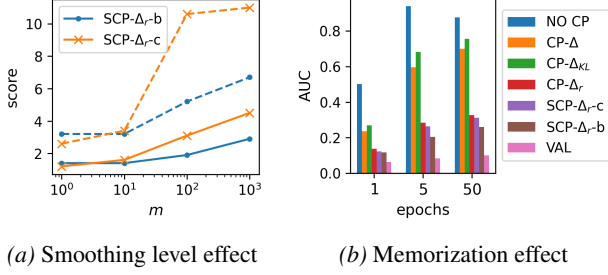


Figure 3. (a) Canary score vs (log scale) smoothing parameter m (mean: solid, 99th percentile: dashed). (b) TTE AUC across different memorization levels on the *MathAbstracts* dataset.

As we increase m , the canary tokens are less likely to be smoothed, and thus protection decreases, which supports Proposition 4.8 (See Appendix C.4 for further discussion). In terms of utility, the only noticeable degradation occurs for $m = 1$, for which both variations achieves a validation accuracy of 0.39. For $m \in \{10, 100, 1000\}$, validation accuracy remains stable at 0.495 ± 0.005 .

Effect of constituent models memorization. Figure 3b shows the effect of the number of training epochs on the TTE attack for the *MathAbstracts* dataset. We focus on tokens that q fails to predict, since tokens that q predicts are trivial to recover and hinder distinguishability between methods for low memorization. Notably, the ranking of the different algorithms by AUC is consistent across memorization levels.

Combining smoothing with raw probabilities. Applying CP- Δ to $\tilde{p}, \tilde{q}$ being raw probability distributions corresponding to the smoothed RPDs $\tilde{r}\tilde{p}, \tilde{r}\tilde{q}$ yields a canary score mean and 95th percentile of 8.1 and 18.3, respectively. These values are comparable to those obtained by CP- Δ (Table 2), indicating that aggregation over RPDs is required for smoothing to achieve a meaningful reduction in extraction.

6. Discussion

Relevance to Pre-Training. The principles underlying SCP- Δ_r may be relevant to pre-training. At the same time, pre-training differs in scale, optimization dynamics and data heterogeneity. We therefore view SCP- Δ_r as a fine-tuning defense, while its extension to pre-training remains an interesting direction for future work.

Assuming Partition Structure. Our guarantees rely on the standard partition assumption used by NAF-based methods: for any sensitive example, at least one constituent model is not trained on that example or its semantic equivalent. This assumption is common in prior work on safe aggregation and copyright protection (Vyas et al., 2023; Abad et al., 2025). Moreover, Abad et al. (2025) showed

that CP-style algorithms can empirically retain a comparable level of protection even when the dataset partitions largely overlap (e.g., up to 33%). Extending NAF-based methods to settings with near-duplicates in the training data remains an important direction for future work; one potential approach is to group semantically similar examples within the same partition.

Reliance on Likelihood-Ratio Optimality Assumptions.

Our extraction bounds do not assume a specific adversarial strategy. Instead, they provide uniform bounds on the likelihood ratio $\frac{r_1(y|x)}{r_0(y|x)}$. This likelihood ratio is the optimal test statistic for membership inference under reasonable assumptions, and also captures the information gain relevant to candidate generation. While the bounds themselves hold independently of how the adversary generates or selects candidates, their interpretation as effective constraints on TDE advantage relies on this connection to optimality. It remains an open question in which regimes these assumptions hold, and whether attacks meaningfully stronger than LiRA exist when they do not.

7. Conclusion

Fine-tuning can amplify memorization and enable training data extraction (TDE) attacks, yet existing defenses either lack formal guarantees or incur substantial utility loss. In particular, CP-style NAF guarantees are often insufficient in realistic fine-tuning settings due to probability spikes.

To address this, we introduced the SpaRPS property to capture sparsity in relative probability shifts and proposed SCP- Δ_r , a NAF-based algorithm that operates on relative probability distributions and enforces SpaRPS via base-model smoothing. Our method achieves orders-of-magnitude stronger TDE guarantees than existing CP-style methods and empirically reduces canary exposure and PII extraction success while preserving utility.

Overall, our results demonstrate that our combination of relative probabilities and targeted smoothing makes NAF-style defenses practically effective for protecting fine-tuned LLMs against training data extraction. In contrast, previously proposed NAF-style defenses are practically vulnerable to our attacks. Extending this approach to relaxed assumptions and broader training regimes remains an important direction for future work.

Impact Statement

This paper presents methods for protecting fine-tuned large language models from training data extraction attacks by providing strong theoretical and empirical privacy guarantees. The primary intended impact of this work is to improve the safety of machine learning systems trained on

proprietary or sensitive data, including personal information, confidential documents, and private codebases.

By reducing the risk of unintended memorization and extraction, our results may help enable more responsible deployment of fine-tuned language models in settings where privacy concerns currently limit their use. Overall, this work advances the technical foundations of privacy-preserving machine learning, and its broader societal implications are largely positive and well understood.

References

- Abad, J., Donhauser, K., Pinto, F., and Yang, F. Copyright-protected language generation via adaptive model fusion. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=kRoWeLTpL4>.
- Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., and Zhang, L. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security, CCS '16*, pp. 308–318, New York, NY, USA, 2016. Association for Computing Machinery. ISBN 9781450341394. doi: 10.1145/2976749.2978318. URL <https://doi.org/10.1145/2976749.2978318>.
- Akkus, A., Aghdam, M. P., Li, M., Chu, J., Backes, M., Zhang, Y., and Sav, S. Generated data with fake privacy: Hidden dangers of fine-tuning large language models on generated data. In *34th USENIX Security Symposium (USENIX Security 25)*, pp. 8075–8093, Seattle, WA, August 2025. USENIX Association. ISBN 978-1-939133-52-6. URL <https://www.usenix.org/conference/usenixsecurity25/presentation/akkus>.
- Blanco-Justicia, A., Sánchez, D., Domingo-Ferrer, J., and Muralidhar, K. A critical review on the use (and misuse) of differential privacy in machine learning. *ACM Comput. Surv.*, 55(8), December 2022. ISSN 0360-0300. doi: 10.1145/3547139. URL <https://doi.org/10.1145/3547139>.
- Carlini, N., Liu, C., Erlingsson, Ú., Kos, J., and Song, D. The secret sharer: Evaluating and testing unintended memorization in neural networks. In *28th USENIX Security Symposium (USENIX Security 19)*, pp. 267–284, Santa Clara, CA, August 2019. USENIX Association. ISBN 978-1-939133-06-9. URL <https://www.usenix.org/conference/usenixsecurity19/presentation/carlini>.
- Carlini, N., Tramèr, F., Wallace, E., Jagielski, M., Herbert-Voss, A., Lee, K., Roberts, A., Brown, T., Song, D., Erlingsson, Ú., Oprea, A., and Raffel, C. Extracting training data from large language models. In *30th USENIX Security Symposium (USENIX Security 21)*, pp. 2633–2650. USENIX Association, August 2021. ISBN 978-1-939133-24-3. URL <https://www.usenix.org/conference/usenixsecurity21/presentation/carlini-extracting>.
- Cohen, A. Blameless users in a clean room: Defining copy-right protection for generative models. In Belgrave, D., Zhang, C., Lin, H., Pascanu, R., Koniusz, P., Ghassemi, M., and Chen, N. (eds.), *Advances in Neural Information Processing Systems*, pp. 56872–56898. Curran Associates, Inc., 2025.
- Du, H., Liu, S., and Cao, Y. Can differentially private fine-tuning llms protect against privacy attacks? In Katsikas, S. and Shafiq, B. (eds.), *Data and Applications Security and Privacy XXXIX*, pp. 311–329, Cham, 2025. Springer Nature Switzerland. ISBN 978-3-031-96590-6.
- Dwork, C. Differential privacy. In Bugliesi, M., Preneel, B., Sassone, V., and Wegener, I. (eds.), *Automata, Languages and Programming*, pp. 1–12, Berlin, Heidelberg, 2006. Springer Berlin Heidelberg. ISBN 978-3-540-35908-1.
- Fan, A., Lewis, M., and Dauphin, Y. Hierarchical neural story generation. In Gurevych, I. and Miyao, Y. (eds.), *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 889–898, Melbourne, Australia, July 2018. Association for Computational Linguistics. doi: 10.18653/v1/P18-1082. URL <https://aclanthology.org/P18-1082/>.
- Fu, W., Wang, H., Gao, C., Liu, G., Li, Y., and Jiang, T. Practical membership inference attacks against fine-tuned large language models via self-prompt calibration. *arXiv preprint arXiv:2311.06062*, 2023.
- Galende, B. A., Apellániz, P. A., Parras, J., Zazo, S., and Uribe, S. Membership inference attacks and differential privacy: A study within the context of generative models. *IEEE Open Journal of the Computer Society*, 6:801–811, 2025. doi: 10.1109/OJCS.2025.3572244.
- Geifman, Y. and El-Yaniv, R. Selective classification for deep neural networks. In *Proceedings of the 31st International Conference on Neural Information Processing Systems, NIPS'17*, pp. 4885–4894, Red Hook, NY, USA, 2017. Curran Associates Inc. ISBN 9781510860964.
- Hans, A., Wen, Y., Jain, N., Kirchenbauer, J., Kazemi, H., Singhanian, P., Singh, S., Somepalli, G., Geiping, J., Bhatele, A., and Goldstein, T. Be like a goldfish, don't memorize! mitigating memorization in generative llms. In *Proceedings of the 38th International Conference on*

- Neural Information Processing Systems, NIPS '24*, Red Hook, NY, USA, 2024. Curran Associates Inc. ISBN 9798331314385.
- Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., and Steinhardt, J. Measuring massive multitask language understanding. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=d7KBjmI3GmQ>.
- J, M. R., VM, K., Warriar, H., and Gupta, Y. Fine tuning llm for enterprise: Practical guidelines and recommendations, 2024. URL <https://arxiv.org/abs/2404.10779>.
- Kalai, A. T., Kalai, Y. T., and Zamir, O. Consensus sampling for safer generative ai. *arXiv preprint arXiv:2511.09493*, 2025.
- Kandpal, N., Wallace, E., and Raffel, C. Deduplicating training data mitigates privacy risks in language models. In Chaudhuri, K., Jegelka, S., Song, L., Szepesvari, C., Niu, G., and Sabato, S. (eds.), *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 10697–10707. PMLR, 17–23 Jul 2022. URL <https://proceedings.mlr.press/v162/kandpal22a.html>.
- Lee, K., Ippolito, D., Nystrom, A., Zhang, C., Eck, D., Callison-Burch, C., and Carlini, N. Deduplicating training data makes language models better. In Muresan, S., Nakov, P., and Villavicencio, A. (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 8424–8445, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.577. URL <https://aclanthology.org/2022.acl-long.577/>.
- Li, R., allal, L. B., Zi, Y., Muennighoff, N., Kocetkov, D., Mou, C., Marone, M., Akiki, C., LI, J., Chim, J., Liu, Q., Zheltonozhskii, E., Zhuo, T. Y., Wang, T., Dehaene, O., Lamy-Poirier, J., Monteiro, J., Gontier, N., Yee, M.-H., Umapathi, L. K., Zhu, J., Lipkin, B., Oblokulov, M., Wang, Z., Murthy, R., Stillerman, J. T., Patel, S. S., Abulkhanov, D., Zocca, M., Dey, M., Zhang, Z., Bhat-tacharyya, U., Yu, W., Luccioni, S., Villegas, P., Zhdanov, F., Lee, T., Timor, N., Ding, J., Schlesinger, C. S., Schoelkopf, H., Ebert, J., Dao, T., Mishra, M., Gu, A., Anderson, C. J., Dolan-Gavitt, B., Contractor, D., Reddy, S., Fried, D., Bahdanau, D., Jernite, Y., Ferrandis, C. M., Hughes, S., Wolf, T., Guha, A., Werra, L. V., and de Vries, H. Starcoder: may the source be with you! *Transactions on Machine Learning Research*, 2023. ISSN 2835-8856. URL <https://openreview.net/forum?id=KoFOg41haE>. Reproducibility Certification.
- Li, T., Liu, Q., Pang, T., Du, C., Guo, Q., Liu, Y., and Lin, M. Purifying large language models by ensembling a small language model, 2024. URL <https://arxiv.org/abs/2402.14845>.
- Lukas, N., Salem, A., Sim, R., Tople, S., Wutschitz, L., and Zanella-Beguelin, S. Analyzing Leakage of Personally Identifiable Information in Language Models . In *2023 IEEE Symposium on Security and Privacy (SP)*, pp. 346–363, Los Alamitos, CA, USA, May 2023. IEEE Computer Society. doi: 10.1109/SP46215.2023.10179300. URL <https://doi.ieeecomputersociety.org/10.1109/SP46215.2023.10179300>.
- Marinelli, R. and Eckhoff, M. Leaking lora: An evaluation of password leaks and knowledge storage in large language models. In Guidotti, R., Schmid, U., and Longo, L. (eds.), *Explainable Artificial Intelligence*, pp. 281–294, Cham, 2026. Springer Nature Switzerland. ISBN 978-3-032-08317-3.
- Mireshghallah, F., Goyal, K., Uniyal, A., Berg-Kirkpatrick, T., and Shokri, R. Quantifying privacy risks of masked language models using membership inference attacks. In Goldberg, Y., Kozareva, Z., and Zhang, Y. (eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 8332–8347, Abu Dhabi, United Arab Emirates, December 2022a. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.570. URL <https://aclanthology.org/2022.emnlp-main.570/>.
- Mireshghallah, F., Uniyal, A., Wang, T., Evans, D., and Berg-Kirkpatrick, T. An empirical analysis of memorization in fine-tuned autoregressive language models. In Goldberg, Y., Kozareva, Z., and Zhang, Y. (eds.), *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 1816–1826, Abu Dhabi, United Arab Emirates, December 2022b. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.119. URL <https://aclanthology.org/2022.emnlp-main.119/>.
- Morales, F. T. Instructional code search net python. https://huggingface.co/datasets/Nan-Do/instructional_code-search-net-python, 2023. Accessed: 2025-12-26.
- Nasr, M., Rando, J., Carlini, N., Hayase, J., Jagielski, M., Cooper, A. F., Ippolito, D., Choquette-Choo, C. A., Tramèr, F., and Lee, K. Scalable extraction of training data from aligned, production language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=vjel3nWP2a>.

- Patil, R. and Gudivada, V. A review of current trends, techniques, and challenges in large language models (llms). *Applied Sciences*, 14(5):2074, 2024.
- Ran, D., He, X., Cong, T., Wang, A., Li, Q., and Wang, X. Lora-leak: Membership inference attacks against lora fine-tuned language models, 2025. URL <https://arxiv.org/abs/2507.18302>.
- Sablayrolles, A., Douze, M., Schmid, C., Ollivier, Y., and Jegou, H. White-box vs black-box: Bayes optimal strategies for membership inference. In Chaudhuri, K. and Salakhutdinov, R. (eds.), *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pp. 5558–5567. PMLR, 09–15 Jun 2019. URL <https://proceedings.mlr.press/v97/sablayrolles19a.html>.
- Segal, T., Shabtai, A., and Elovici, Y. Domba: Double model balancing for access-controlled language models via minimum-bounded aggregation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 39, pp. 25101–25109, 2025.
- Tirumala, K., Markosyan, A., Zettlemoyer, L., and Aghajanyan, A. Memorization without overfitting: Analyzing the training dynamics of large language models. *Advances in Neural Information Processing Systems*, 35: 38274–38290, 2022.
- Touvron, H., Martin, L., Stone, K., Albert, P., Almahairi, A., Babaei, Y., Bashlykov, N., Batra, S., Bhargava, P., Bhosale, S., Bikel, D., Blecher, L., Ferrer, C. C., Chen, M., Cucurull, G., Esiobu, D., Fernandes, J., Fu, J., Fu, W., Fuller, B., Gao, C., Goswami, V., Goyal, N., Hartshorn, A., Hosseini, S., Hou, R., Inan, H., Kardas, M., Kerkez, V., Khabsa, M., Kloumann, I., Korenev, A., Koura, P. S., Lachaux, M.-A., Lavril, T., Lee, J., Liskovich, D., Lu, Y., Mao, Y., Martinet, X., Mihaylov, T., Mishra, P., Molybog, I., Nie, Y., Poulton, A., Reizenstein, J., Rungta, R., Saladi, K., Schelten, A., Silva, R., Smith, E. M., Subramanian, R., Tan, X. E., Tang, B., Taylor, R., Williams, A., Kuan, J. X., Xu, P., Yan, Z., Zarov, I., Zhang, Y., Fan, A., Kambadur, M., Narang, S., Rodriguez, A., Stojnic, R., Edunov, S., and Scialom, T. Llama 2: Open foundation and fine-tuned chat models, 2023. URL <https://arxiv.org/abs/2307.09288>.
- Vyas, N., Kakade, S. M., and Barak, B. On provable copyright protection for generative models. In *International conference on machine learning*, pp. 35277–35299. PMLR, 2023.
- Zhang, J., Sun, J., Yeats, E., Ouyang, Y., Kuo, M., Zhang, J., Yang, H. F., and Li, H. Min-k%++: Improved baseline for pre-training data detection from large language models. In *The Thirteenth International Conference on Learning Representations*, 2025a.
- Zhang, Y., Luo, Y., Yuan, Y., and Yao, A. C. Autonomous data selection with zero-shot generative classifiers for mathematical texts. In Che, W., Nabende, J., Shutova, E., and Pilehvar, M. T. (eds.), *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 4168–4189, Vienna, Austria, July 2025b. Association for Computational Linguistics. ISBN 979-8-89176-256-5. doi: 10.18653/v1/2025.findings-acl.216. URL <https://aclanthology.org/2025.findings-acl.216/>.
- Zhou, J., Lu, T., Mishra, S., Brahma, S., Basu, S., Luan, Y., Zhou, D., and Hou, L. Instruction-following evaluation for large language models. *arXiv preprint arXiv:2311.07911*, 2023.

A. Bounding SCP- Δ_r Utility Degradation

For $r = \text{CP-}\Delta(p, q)$, Vyas et al. (2023) proved that $TV(r, q) \leq TV(q, p)$ (and by symmetry, $TV(r, p) \leq TV(q, p)$). This implies that if p and q are similar in total variation, then r is also close to both, and therefore its utility should be comparable.

For SCP- Δ_r , utility degradation may arise from two sources: (i) the smoothing mechanism, and (ii) the aggregation of relative probability distributions (RPDs). In the following, we derive separate bounds for each source of potential degradation, beginning with the aggregation stage.

Lemma A.1 (Utility degradation bound - aggregation). *Let $r = \text{CP-}\Delta_r(p, q)$. Then*

$$TV(r, q) \leq TV(p, q).$$

Proof is provided in Appendix B.7.

Lemma A.2 (Utility degradation bound - smoothing). *Let $\tilde{p}$ be the probability distributions corresponding to the RPD $\tilde{r}\tilde{p} = S_m(rp, rb)$ (see Algorithm 1) and A be a set of m “unsmoothed” tokens and $\bar{A}$ its complement. Let $\epsilon = \frac{tp}{tb} \sum_{y \in \bar{A}} b(y)$. Then*

$$\Delta_{\text{KL}}(p \| \tilde{p}) \leq \log(1 + \epsilon) + \sum_{y \in \bar{A}} p(y) \log \frac{rp(y)}{rb(y)}.$$

Consequently, by Pinsker’s inequality, $TV(p, \tilde{p}) \leq \sqrt{\frac{1}{2} \Delta_{\text{KL}}(p \| \tilde{p})}$.

Proof is provided in Appendix B.8.

Remark A.3. The smoothing operator selects the set A as

$$A \in \arg \max_{|A|=m} \sum_{y \in A} p(y) \log \frac{rp(y)}{rb(y)}.$$

Equivalently, since $\sum_y p(y) \log \frac{rp(y)}{rb(y)}$ is constant with respect to A , this is the same as

$$A \in \arg \min_{|A|=m} \sum_{y \in \bar{A}} p(y) \log \frac{rp(y)}{rb(y)}.$$

Thus, A is the optimal choice of size m for minimizing the upper bound on $\Delta_{\text{KL}}(p \| \tilde{p})$ given in Lemma A.2.

Remark A.4. Since p is a fine-tuned model and b is a base model, we typically expect $tp \lesssim tb$ (more probability spikes implies lower typical probability). There are two common regimes in which $\epsilon \ll 1$.

First, if $\sum_{y \in \bar{A}} b(y) \ll 1$, which occurs when b is close to p and the unsmoothed m tokens are sufficient to capture most of the probability mass.

Second, if $tp \ll tb$, which occurs when p exhibits probability spikes that concentrate mass on a small number of tokens, thereby reducing the mass assigned to typical tokens.

B. Proofs

B.1. Proof of Theorem 3.1

Proof. Let $a_y = \log \frac{p(y)}{q(y)}$. Then

$$\sum_y \log^2 \left(C \frac{p(y)}{q(y)} \right) = \sum_y (\log C + a_y)^2.$$

It is well known that the minimizer of $\sum_i (x + a_i)^2$ over x is $x = -\frac{1}{n} \sum_i a_i$. Applying this yields

$$\log C = -\frac{1}{n} \sum_y \log \frac{p(y)}{q(y)} = \log \frac{tq}{tp},$$

and therefore $C = \frac{tq}{tp}$. □

B.2. Proof of Proposition 3.3

Proof. Let $A \subseteq Y$ satisfy $|A| = m$ and $p(y) = \alpha q(y)$ for all $y \in Y \setminus A$, with $\alpha > 0$. For $y \notin A$ we have $\log q(y) - \log p(y) = -\log \alpha$, and therefore

$$\log \frac{\alpha tq}{tp} = \log \alpha + \frac{1}{n} \sum_{y \in Y} (\log q(y) - \log p(y)) = \frac{m}{n} \log \alpha + \frac{1}{n} \sum_{y \in A} (\log q(y) - \log p(y)).$$

Hence, using $\log \frac{p(y)}{q(y)} \leq \log M$ for all $y \in A$,

$$\log \frac{\alpha tq}{tp} \leq \frac{m}{n} |\log \alpha| + \frac{1}{n} \sum_{y \in A} \log \frac{q(y)}{p(y)} \leq \frac{m}{n} (|\log \alpha| + \log M).$$

Next,

$$D_r(p, q) = \frac{1}{2n} \sum_{y \in Y} \log \frac{p(y)}{\alpha q(y)} + \log \frac{\alpha tq}{tp} \leq \frac{1}{2n} \sum_{y \in Y} \log \frac{p(y)}{\alpha q(y)} + \frac{1}{2} \log \frac{\alpha tq}{tp}.$$

For $y \notin A$, $\log \frac{p(y)}{\alpha q(y)} = 0$, and for $y \in A$,

$$\log \frac{p(y)}{\alpha q(y)} = \log \frac{p(y)}{q(y)} - \log \alpha \leq \log \frac{p(y)}{q(y)} + |\log \alpha| \leq \log M + |\log \alpha|.$$

Thus,

$$\sum_{y \in Y} \log \frac{p(y)}{\alpha q(y)} = \sum_{y \in A} \log \frac{p(y)}{\alpha q(y)} \leq m (\log M + |\log \alpha|).$$

Combining the bounds yields

$$D_r(p, q) \leq \frac{m}{2n} (\log M + |\log \alpha|) + \frac{1}{2} \cdot \frac{m}{n} (\log M + |\log \alpha|) = \frac{m}{n} (\log M + |\log \alpha|).$$

□

B.3. Proof of Remark 4.1

Proof. Let $sp = S_m(rp, rb)$ and $sq = S_m(rq, rb)$. Let $A_p \subseteq Y$ denote the set of indices whose outputs in sp originate from rp , and define $A_q \subseteq Y$ analogously for sq . By construction of the smoothing operator, $|A_p| = |A_q| = m$.

For all $y \in Y \setminus A_p$, there exists a constant $\alpha_p > 0$ such that $sp(y) = \alpha_p rb(y)$. Similarly, for all $y \in Y \setminus A_q$, there exists a constant $\alpha_q > 0$ such that $sq(y) = \alpha_q rb(y)$. Therefore, for all $y \in Y \setminus (A_p \cup A_q)$,

$$sp(y) = \alpha_p rb(y) = \frac{\alpha_p}{\alpha_q} sq(y).$$

Let $A := A_p \cup A_q$. Since $|A| \leq 2m$, if $|A| < 2m$ we enlarge A arbitrarily to a superset $A' \supseteq A$ with $|A'| = 2m$. The proportionality above continues to hold for all $y \in Y \setminus A'$. Hence sp is $2m$ -SpaRPS with respect to sq . □

B.4. Proof of Theorem 4.4

.

Proof. Let $rr_1 := \text{RPD}(r_1)$ and $rq := \text{RPD}(q)$, i.e., $rr_1(y|x) = r_1(y|x)/tr_1(x)$ and $rq(y|x) = q(y|x)/tq(x)$. Then for any y ,

$$\frac{r_1(y|x)}{q(y|x)} = \frac{rr_1(y|x)}{rq(y|x)} \cdot \frac{tr_1(x)}{tq(x)} \leq e^{k_x} \cdot \frac{tr_1(x)}{tq(x)}.$$

□

B.5. Proof of Theorem 4.5

Proof. By definition of CP- Δ_r , for $i \in \{0, 1\}$,

$$r_i(y) = \frac{\min(rp_i(y), rq(y))}{Z_i}, \quad Z_i = \sum_y \min(rp_i(y), rq(y)).$$

Hence,

$$\frac{r_1(y)}{r_0(y)} = \frac{\min(rp_1(y), rq(y))}{\min(rp_0(y), rq(y))} \cdot \frac{Z_0}{Z_1}.$$

Since $\min(rp_1(y), rq(y)) \leq rq(y)$,

$$\frac{r_1(y)}{r_0(y)} \leq \frac{rq(y)}{\min(rp_0(y), rq(y))} \cdot \frac{Z_0}{Z_1} = \max\left(1, \frac{rq(y)}{rp_0(y)}\right) \cdot \frac{Z_0}{Z_1}.$$

Let $rr_i := \text{RPD}(r_i) = r_i/tr_i$. Then

$$rr_i(y) = \frac{r_i(y)}{tr_i} = \frac{\min(rp_i(y), rq(y))}{Z_i tr_i},$$

so $\frac{rr_i(y)}{\min(rp_i(y), rq(y))} = \frac{1}{Z_i tr_i}$ is constant in y . By the tight form of Theorem 2.7 (i.e., the maximum is attained as an equality),

$$\max_y \frac{rr_i(y)}{\min(rp_i(y), rq(y))} = e^{D_r(p_i, q)},$$

and therefore (since the ratio is constant in y)

$$\frac{1}{Z_i tr_i} = e^{D_r(p_i, q)} \implies Z_i = \frac{e^{-D_r(p_i, q)}}{tr_i}.$$

Thus,

$$\frac{Z_0}{Z_1} = \frac{tr_1}{tr_0} \cdot e^{D_r(p_1, q) - D_r(p_0, q)} \leq \frac{tr_1}{tr_0} \cdot e^{k_x},$$

where $k_x := D_r(p_1, q)$ and $D_r(p_0, q) \geq 0$. Substituting back completes the proof. $\square$

B.6. Proof of Proposition 4.8

Proof. Let b be the base model and let $\tilde{r}q = S_m(rq, rb)$ and $\tilde{r}p_0 = S_m(rp_0, rb)$ denote the smoothed RPDs. By assumption, y is smoothed in both, hence

$$\tilde{r}q(y|x) = \beta_q(x) rb(y|x), \quad \tilde{r}p_0(y|x) = \beta_0(x) rb(y|x),$$

for some normalization constants $\beta_q(x), \beta_0(x) > 0$ chosen so that the resulting distributions are valid RPDs (i.e., have zero mean log).

Let $A_q(x)$ (resp. $A_0(x)$) be the unsmoothed token set of size m for q (resp. p_0) at input x . From the RPD normalization constraint $\sum_{z \in Y} \log \tilde{r}q(z|x) = 0$ and $\sum_{z \in Y} \log rb(z|x) = 0$, we obtain

$$n \log \beta_q(x) = \sum_{z \in A_q(x)} \left(\log rb(z|x) - \log rq(z|x) \right),$$

and similarly,

$$n \log \beta_0(x) = \sum_{z \in A_0(x)} \left(\log rb(z|x) - \log rp_0(z|x) \right),$$

where $n := |Y|$.

Assume a bounded log-ratio condition on the unsmoothed tokens, namely that for all relevant x and all $z \in A_q(x)$, $|\log rb(z|x) - \log rq(z|x)| \leq B$, and analogously for $A_0(x)$. Then

$$|\log \beta_q(x)| \leq \frac{mB}{n}, \quad |\log \beta_0(x)| \leq \frac{mB}{n}.$$

Therefore,

$$\frac{\tilde{r}q(y|x)}{\tilde{r}p_0(y|x)} = \frac{\beta_q(x) rb(y|x)}{\beta_0(x) rb(y|x)} = \exp(\log \beta_q(x) - \log \beta_0(x)) \leq \exp(|\log \beta_q(x)| + |\log \beta_0(x)|) \leq \exp\left(\frac{2mB}{n}\right).$$

Note that since all probabilities are lower bounded by e^{-20} , all typical probabilities are also lower bounded by e^{-20} . Consequently, all relative probabilities lie in the range $[e^{-20}, e^{20}]$, implying $B = 40$. Plugging in $m = 10$ and $n = 32000$ gives $\exp(2mB/n) = \exp(800/32000) < 1.026$, completing the proof. $\square$

B.7. Proof of Lemma A.1

Proof. For distributions u, v , $TV(u, v) = 1 - \sum_y \min(u(y), v(y))$. Write

$$r(y) = \frac{\min\left(\frac{p(y)}{tp}, \frac{q(y)}{tq}\right)}{z} = \min\left(\frac{p(y)}{ztp}, \frac{q(y)}{ztq}\right), \quad z = \sum_y \min\left(\frac{p(y)}{tp}, \frac{q(y)}{tq}\right).$$

Note that

$$ztp = \sum_y \min\left(p(y), q(y) \frac{tp}{tq}\right) \leq \sum_y p(y) = 1, \quad ztq = \sum_y \min\left(p(y) \frac{tq}{tp}, q(y)\right) \leq \sum_y q(y) = 1.$$

Hence $\frac{p(y)}{ztp} \geq p(y)$ and $\frac{q(y)}{ztq} \geq q(y)$ for all y . Let $A_y = \frac{p(y)}{ztp}$ and $B_y = \frac{q(y)}{ztq}$. Then $r(y) = \min(A_y, B_y)$ and $B_y \geq q(y)$, so

$$\min(r(y), q(y)) = \min(A_y, B_y, q(y)) = \min(A_y, q(y)) \geq \min(p(y), q(y)).$$

Summing over y yields $\sum_y \min(r(y), q(y)) \geq \sum_y \min(p(y), q(y))$, hence $TV(r, q) \leq TV(p, q)$. $\square$

B.8. Proof of Lemma A.2

Proof. Let $rp(y) = p(y)/tp$ and $rb(y) = b(y)/tb$ with $tp, tb > 0$. Let A be a set of m “unsmoothed” tokens and $\bar{A}$ its complement. Define the smoothed RPD

$$\tilde{r}p(y) = \beta \begin{cases} rp(y) & y \in A, \\ rb(y) & y \in \bar{A}, \end{cases}$$

where $\beta > 0$ normalizes $\tilde{r}p$ into an RPD. After converting back to probabilities: $\tilde{p}(y) = \tilde{r}p(y) \tilde{tp}$ for some $\tilde{tp} \geq 0$ (so that $\sum_y \tilde{p}(y) = 1$). Let $\epsilon = \frac{tp}{tb} \sum_{y \in \bar{A}} b(y)$. Since $\sum_y \tilde{p}(y) = 1$ and $\tilde{p}(y) = \beta \tilde{tp} rp(y)$ for $y \in A$ and $\tilde{p}(y) = \beta \tilde{tp} rb(y)$ for $y \in \bar{A}$, we have

$$1 = \beta \tilde{tp} \left(\sum_{y \in A} rp(y) + \sum_{y \in \bar{A}} rb(y) \right) \Rightarrow \frac{1}{\beta \tilde{tp}} = \sum_{y \in A} rp(y) + \sum_{y \in \bar{A}} rb(y).$$

Rewrite the RHS using $rp = p/tp$ and $rb = b/tb$:

$$\frac{1}{\beta \tilde{tp}} = \frac{1}{tp} \sum_{y \in A} p(y) + \frac{1}{tb} \sum_{y \in \bar{A}} b(y) \leq \frac{1}{tp} \left(1 + \frac{tp}{tb} \sum_{y \in \bar{A}} b(y) \right) = \frac{1 + \epsilon}{tp}.$$

Hence $\beta \tilde{tp} \geq \frac{tp}{1 + \epsilon}$.

If $y \in A$ then

$$\tilde{p}(y) = \beta \tilde{tp} rp(y) \geq \frac{tp}{1 + \epsilon} \cdot \frac{p(y)}{tp} = \frac{p(y)}{1 + \epsilon},$$

so $\log \frac{p(y)}{\tilde{p}(y)} \leq \log(1 + \epsilon)$.

If $y \in \bar{A}$ then

$$\tilde{p}(y) = \beta \tilde{t} p r b(y) \geq \frac{tp}{1 + \epsilon} \cdot \frac{b(y)}{tb},$$

so

$$\frac{p(y)}{\tilde{p}(y)} \leq (1 + \epsilon) \frac{p(y)tb}{b(y)tp} = (1 + \epsilon) \frac{rp(y)}{rb(y)},$$

and thus $\log \frac{p(y)}{\tilde{p}(y)} \leq \log(1 + \epsilon) + \log \frac{rp(y)}{rb(y)}$.

Summing over A and $\bar{A}$ yields

$$\Delta_{\text{KL}}(p \parallel \tilde{p}) \leq \log(1 + \epsilon) + \sum_{y \in \bar{A}} p(y) \log \frac{rp(y)}{rb(y)}.$$

Pinsker gives the stated TV bound. □

C. Further discussions

C.1. A TDE Optimal Strategy

The likelihood-ratio attack (LiRA) has been shown to be optimal for membership inference (MI) under a set of standard assumptions. We state these assumptions informally below; for precise formulations and proofs, see (Sablayrolles et al., 2019):

1. The distribution over model parameters θ is assumed to follow a fixed likelihood model determined by the training loss induced by θ .
2. The reference model r_0 is trained using the same procedure and on data drawn from a distribution similar to that of r_1 .
3. Candidate members (x_i, y_i) are drawn i.i.d. from the data distribution, each having probability λ of being a member of the training set.

While Assumption 3 does not generally hold for all TDE attacks, LiRA can nevertheless remain relevant even when this assumption is violated; see Appendix C.2.

C.2. Relevance of LiRA to Our Attacks

Our attacks do not explicitly rely on a reference model and are therefore potentially weaker than the optimal likelihood-ratio attack (LiRA) analyzed in our theory. Nevertheless, LiRA remains closely connected to our attack mechanisms. Below, we clarify this connection for each attack setting.

Canary method. Our canaries consist of sequences of three random words. An ideal reference model would assign nearly uniform probability across such sequences, substantially limiting the usefulness of an explicit reference model. In this case, LiRA effectively reduces to thresholding the target model’s output probabilities, making reference-model-free attacks comparable in strength.

PII extraction. For PII extraction, an ideal reference model would assign uniform probability to all digits. In this case $r_0(y)$ is constant over candidate digits, so the likelihood ratio $\frac{r_1(y|x)}{r_0(y|x)}$ differs from $r_1(y|x)$ only by a constant scaling. Thus, using LiRA is equivalent to thresholding $r_1(y|x)$ after adjusting the decision threshold.

Token-by-token extraction (TTE). In TTE, candidate tokens are not drawn from a fixed pool but arise naturally from the language model’s generation process. As a result, some of the assumptions under which LiRA is optimal for the membership inference stage do not strictly hold, in particular Assumption 3 in Appendix C.1. For example, given the prefix “12345,” the next token “6” is highly likely due to language structure alone, even if the likelihood ratio between the target and reference models is small. In such cases, thresholding the target model probability can be more appropriate than directly applying

LiRA. Nevertheless, as discussed in Section 4.2, the likelihood ratio still quantifies the information gain an adversary obtains from observing the target model’s output. Bounding this ratio therefore constrains the adversary’s advantage in the candidate generation stage, which makes LiRA-based analysis relevant for TTE as well.

C.3. Choice of Smoothing Parameter m

We did not perform hyperparameter tuning to select the smoothing parameter m . Instead, we fix $m = 10$ throughout all experiments except in the ablation study that examines the effect of varying m . This choice reflects a trade-off between expressiveness and robustness: on the one hand, m is small relative to the vocabulary size n , which is typically on the order of tens of thousands of tokens; on the other hand, retaining ten tokens appears sufficient to preserve the dominant contributors to the model utility in practice, as validated by our utility metric.

While performance may be further improved by tuning m for a specific dataset or training regime, our results suggests that $m = 10$ serves as a reasonable default that does not rely on dataset-specific or training-specific details.

C.4. Theoretical Analysis of SCP- Δ_r -b vs. SCP- Δ_r -c

As shown in Table 2, SCP- Δ_r -b and SCP- Δ_r -c achieve comparable performance on the canary method and TTE, while SCP- Δ_r -b provides stronger protection on PII extraction. This difference can be explained by Proposition 4.8.

By the proposition, SCP- Δ_r achieves its strongest advantage over CP- Δ_r on tokens that are smoothed in both $\tilde{r}q$ and $\tilde{r}p_0$. In the PII setting, the tokens of interest are the digits 0-9. Recall that the smoothing operator retains the m tokens with the highest relative information gain relative to the base model and smooths the remaining tokens.

For SCP- Δ_r -b, the base model is the pre-fine-tuning LLM, which is context-aware. As a result, for digit tokens d , it is common that $r_q(d) \approx r_b(d)$, leading to low relative information gain. Consequently, digit tokens are likely to be smoothed, yielding stronger protection.

In contrast, SCP- Δ_r -c uses a non-contextual base model that assigns a constant distribution. Such a base model assigns relatively low probability to digit tokens compared to a context-aware distribution, resulting in high relative information gain for digits. These tokens are therefore less likely to be smoothed, weakening protection on PII extraction.

Finally, note that this reasoning applies when the number of relevant tokens is on the order of or smaller than m . In the canary setting, however, the number of candidate tokens is much larger than m , making the probability that any specific target token is smoothed high regardless of whether the base model is contextual. As illustrated in Figure 3a, increasing m makes this effect apparent for the canary method as well.

D. Additional Training parameters and Experiments

D.1. Training Details

We follow the training setup of Abad et al. (2025). Our default training hyperparameters are reported in Table 3. We use a batch size of 1, since training examples are concatenated with an EOS (end-of-sequence) token up to a maximum sequence length of 2048.

The fine-tuning process requires adding a separator token and a padding token to the tokenizer. We found that the default Hugging Face initialization for newly added tokens led to gradient explosions in our setting. Instead, we initialize the embeddings of new tokens using the EOS token embedding, which we found to be stable.

The LoRA training parameters are reported in Table 3. For LoRA fine-tuning, concatenating training examples dramatically reduced memorization in the constituent models, preventing meaningful evaluation. We therefore use a sequence length of 25, corresponding to the maximum length of examples in the *Names2IDs* dataset (shorter examples are padded with EOS). Additionally, we observed that adding new tokens interacts poorly with LoRA fine-tuning; as a result, we removed the separator token from the dataset and did not introduce new tokens into the tokenizer. Finally, training for 50 epochs led to gradient instability in some runs, so we reduced the number of epochs to 20, which still resulted in substantial memorization.

Table 3. Default training parameters for regular fine-tuning and LoRA.

Parameter	Regular Fine-Tuning Default	LoRA Default
#train sample per partition	3000	3000
epochs	50	20
batch_size	1	64
neptune_noise_alpha	5	None
learning_rate	5e-5	5e-4
gradient_accumulation_steps	1	1
optim	adamw_bnb_8bit	adamw_bnb_8bit
warmup_steps	50	50
max_grad_norm	2.0	2.0
lora_r	-	64
lora_alpha	-	32
lora_dropout	-	0.05
target_modules	-	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj, lm_head

 Table 4. 99th percentile k_x values for the *MathAbstracts* (M), *StoryTelling* (S) and *CodeInstructions* (C) datasets for different CP-style algorithms.

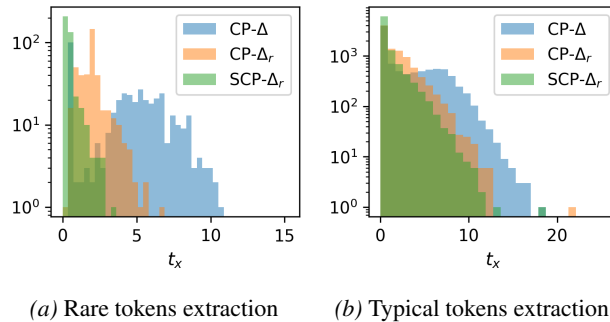
	k_x (99%) ↓		
	M	S	C
CP- Δ	11.6	12.6	14.0
CP- Δ_r	.98	1.58	.84
SCP- Δ_r -b	.004	.005	.003
SCP- Δ_r -c	<u>.005</u>	<u>.008</u>	<u>.004</u>

D.2. k_x Bounds

Table 4 reports the 99th percentile of the k_x bound on the *MathAbstracts*, *StoryTelling*, and *CodeInstructions* datasets. SCP- Δ_r achieves substantially smaller k_x values than CP- Δ_r , which in turn outperforms CP- Δ . The difference between SCP- Δ_r -b and SCP- Δ_r -c is minor, with a slight advantage for SCP- Δ_r -b.

D.3. t_x Bounds

Figure 4 presents empirical histograms of the t_x bounds (Theorem 4.4) for different CP-style algorithms. SCP- Δ_r achieves the smallest t_x bounds, followed by CP- Δ_r .


 Figure 4. Log-scale histograms of Theorem 4.4 bound t_x . (a) Extraction of rare tokens (Canaries, see 5.3). (b) Extraction of tokens from the *MathAbstracts* dataset (see 5.1).

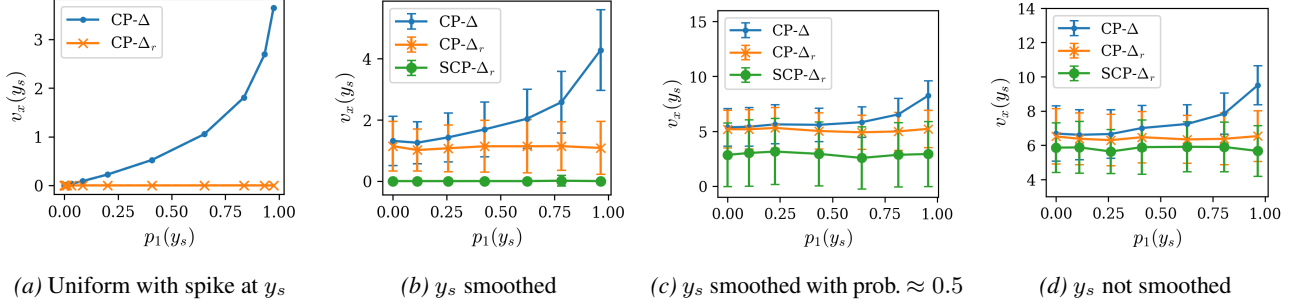


Figure 5. Theorem 4.5 bound $v_x(y_s)$ as a function of $p_1(y_s)$ under different smoothing conditions. Error bars denote one standard deviation. (a) q and p_0 are uniform, while p_1 is uniform except for a spike at y_s . (b) q , p_0 , p_1 and b are randomly initialized, with p_1 additionally spiked at y_s . (c) Same as (b), but $q(y_s)$ is scaled so that y_s is smoothed with probability approximately 0.5. (d) Same as (b), but $q(y_s) = \max_y q(y)$, making y_s very unlikely to be smoothed.

D.4. Illustration of Proposition 4.8

Proposition 4.8 implies that the bound $v_x(y)$ from Theorem 4.5 improves when the target token y is smoothed. To illustrate this effect, we conduct the following synthetic experiment.

We fix a target token y_s and initialize distributions p_0 , p_1 , q , and the base model distribution b . We then gradually increase $p_1(y_s)$ and measure the resulting bound $v_x(y_s)$ for CP- Δ , CP- Δ_r , and SCP- Δ_r . Different initialization schemes are used to control the likelihood that y_s is smoothed. We also include a simple baseline where all distributions are initialized to be uniform (Figure 5a), in which case CP- Δ_r and SCP- Δ_r coincide.

For the non-uniform settings, we first sample a shared logits vector $w \sim \mathcal{N}(0, 1)$ to induce correlation between models, and then initialize each distribution $p \in \{p_0, p_1, q, b\}$ as $p = \text{Softmax}(w + \mathcal{N}(0, 1))$. We then modify $q(y_s)$ to control whether y_s is smoothed:

Smoothed (Figure 5b): $q(y_s)$ is left unchanged. Since $m \ll n$, the probability that y_s is smoothed is approximately $(n - m)/n$, which is close to 1.

Smoothed with probability ≈ 0.5 (Figure 5c): We multiply $q(y_s)$ by 4.5, which empirically results in y_s being smoothed with probability close to 0.5 for $m = 10$ and $n = 32000$.

Not smoothed (Figure 5d): We set $q(y_s) = \max_y q(y)$, making y_s among the highest relative information-gain tokens and therefore very unlikely to be smoothed.

As shown in Figure 5, CP- Δ_r consistently achieves smaller bounds than CP- Δ , and SCP- Δ_r further improves upon CP- Δ_r . Moreover, the improvement of SCP- Δ_r becomes more pronounced as the likelihood of y_s being smoothed increases, in accordance with Proposition 4.8. Error bars denote standard deviation (not standard error); results are averaged over 200 trials, and all differences between SCP- Δ_r and CP- Δ_r are statistically significant ($p < 10^{-4}$).

Table 5. Canary attack results. Best values are in bold. Underlined values are within two standard deviations of the random-canary baseline (i.e., not significantly above chance). Random-canary baseline: exposure computed for canaries of the same format that were not inserted into training.

	REP = 1		REP = 3		REP = 10	
	Mean ↓	95% ↓	Mean ↓	95% ↓	Mean ↓	95% ↓
No CP	24	28	28	31	29	33
CP- Δ	5.2	13	8.8	17	11	20
CP- Δ_{KL}	16	25	23	33	25	33
CP-Fuse	2.7	5.5	3.8	7.1	4.2	7.2
CP- Δ_r	2.6	6.4	3.6	8	3.8	8.9
SCP- Δ -c	1.5	4.4	<u>1.6</u>	<u>3.4</u>	1.8	<u>4.1</u>
SCP- Δ -b	1.7	<u>4.8</u>	1.4	3.2	1.6	3.7

Table 6. PII average extraction length results with LoRA fine-tuning.

	AEL
No CP	9.79 ± 0.08
CP- Δ	2.6 ± 0.8
CP- Δ_{KL}	5.0 ± 1.3
CP-Fuse	5.0 ± 1.3
CP- Δ_r	0.9 ± 0.23
SCP- Δ -c	0.81 ± 0.07
SCP- Δ -b	0.39 ± 0.05

D.5. The Canary Method

Table 5 presents the full canary attack results on the *MathAbstracts* dataset. Results are qualitatively similar across different numbers of canary repetitions, with higher repetition leading to increased memorization for all methods except for both variations of SCP- Δ_r . Notably, both variations of SCP- Δ_r remain within two standard deviations of the random canary baseline in 5 out of 6 evaluated settings.

D.6. PII Attacks on LoRA Fine-Tuned LLMs

Table 6 reports the average extracted length (AEL) for the PII attack on the *Names2IDs* dataset when fine-tuning with LoRA for 20 epochs. Results are averaged over five trials with different training examples. Notably, LoRA did not prevent memorization, and these results are qualitatively similar to those obtained under non-LoRA fine-tuning (Table 2).

D.7. Reasoning and Instruction-Following Evaluation

In addition to our empirical next-token accuracy evaluation and the KL-divergence bounds introduced by smoothing (Lemma A.2), we evaluate the impact of smoothing on downstream capabilities. We measure reasoning performance on MMLU (5-shot) (Hendrycks et al., 2021) and instruction-following performance on IFEval (Zhou et al., 2023).

For the MMLU evaluation, we use Llama-2-7b, while for the IFEval evaluation we use Llama-2-7b-chat. In both experiments, we set the smoothing parameter to $m = 10$, as in our main experiments. Table 7 reports the results.

The results are consistent with our next-token accuracy findings, demonstrating that smoothing preserves model utility.

Table 7. Utility evaluation with and without smoothing.

Configuration	MMLU Accuracy (5-shot)	IFEval Prompt-Level Accuracy
No Smoothing	45.83%	31.42%
With Smoothing	45.75%	31.61%